MIL-P-46474B (PA)
18 March 1976
SUPERSEDING
MIL-P-46474A (MU)
20 December 1963

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M58
METAL PARTS FOR

This specification is approved for use by Picatinny Arsenal Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for one type of percussion primer designated as Primer, Percussion, M58.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169 - Packaging, packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

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DRAWINGS

ARMY

8838012 - Cup, Battery
8838015 - Plug, Firing
8838085 - Head
8838089 - Body (Part No. 8838089-18)

(Copies of specifications, standard, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.-The metal parts shall comply with all requirements specified on Drawings (dwg.) 8838012, 8838015, 8838085 and 8838089, as applicable, and with all requirements specified in applicable specifications.

3.3 Hydrostatic test.-The tubing to be fabricated into primer bodies shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks and without permanent deformation beyond drawing dimensions when tested as specified in 4.5.1 (see 6.3).

3.4 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, cracks, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

3.5 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance.- Unless otherwise specified herein or in the contract,

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The Provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspection.-The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 25 each of every component which have been produced by the contractor (or furnished to him by a supplier).

4.3.2 Inspection to be performed.-The first Article parts shall be submitted in accordance with paragraph 4.3.1, will be subjected by the Government to any or all the requirements of the applicable drawings and to the inspection in 4.4.

4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46474B (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Cup Battery		8838012
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Outside diameter, minimum (min)		.40%	3.2	Gage
102.	Thickness through bottom, maximum (max)		.40%	3.2	Gage
103.	Length		.40%	3.2	Gage
Minor					
201.	Diameter of hole		.65%	3.2	Gage
202.	Thickness through bottom, min.		.65%	3.2	Gage
203.	Inside diameter		.65%	3.2	Gage
204.	Outside diameter, max.		.65%	3.2	Gage
205.	Flat missing from surface at open end		.65%	3.2	Visual
206.	Evidence of poor workmanship		.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46474B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Plug, Firing				8838015
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Large diameter, min.		.40%	3.2	Gage
102.	Length and profile of point		.40%	3.2	Gage
103.	Length of pilot, min.		.40%	3.2	Gage
104.	Total length		.40%	3.2	Gage
Minor					
201.	Large diameter, max.		.65%	3.2	Gage
202.	Length of pilot, max.		.65%	3.2	Gage
203.	Diameter of pilot		.65%	3.2	Gage
204.	Finish improper		.65%	3.2	Visual
205.	Evidence of poor workmanship		.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46474B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Head				8838085
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Pitch diameter of body thread		.40%	3.2	Gage
102.	Major diameter of body thread, min.		.40%	3.2	Gage
103.	Pitch diameter of case thread		.40%	3.2	Gage
104.	Major diameter of case thread		.40%	3.2	Gage
105.	Diameter of body pilot, max.		.40%	3.2	Gage
106.	Diameter of battery cup cavity		.40%	3.2	Gage
107.	Diameter of percussion element cavity		.40%	3.2	Gage
108.	Depth of percussion element cavity		.40%	3.2	Gage
109.	Thickness of flange, max.		.40%	3.2	Gage
110.	Crack or split		.40%	3.2	Visual
111.	Flash hole missing or obstructed		.40%	3.2	Visual
Minor					
201.	Major diameter of body thread, max.		.65%	3.2	Gage
202.	Diameter of flange		.65%	3.2	Gage
203.	Depth of battery cup cavity		.65%	3.2	Gage
204.	Diameter of body pilot, min.		.65%	3.2	Gage
205.	Concentricity between percussion element cavity and flash hole		.65%	3.2	Gage
206.	Concentricity of percussion element cavity with case thread		.65%	3.2	Gage
207.	Concentricity of flange with case thread		.65%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.3	CONTINUATION SHEET		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
208.	Diameter of chamfer at mouth of percussion element cavity		.65%	3.2	Gage
209.	Diameter of flash hole		.65%	3.2	Gage
210.	Length to shoulder above body thread		.65%	3.2	Gage
211.	Length of body thread undercut, max.		.65%	3.2	Gage
212.	Length of case thread undercut, max.		.65%	3.2	Gage
213.	Length from rear face to case thread shoulder		.65%	3.2	Gage
214.	Diameter of shoulder between threads		.65%	3.2	Gage
215.	Distance between wrench slots		.65%	3.2	Gage
216.	Depth of wrench slots, min.		.65%	3.2	Gage
217.	Total length		.65%	3.2	Gage
218.	Depth of flash cavity		.65%	3.2	Gage
219.	Diameter and profile of flash cavity		.65%	3.2	Gage
220.	Length of body thread including thread undercut		.65%	3.2	Gage
221.	Thread undercut missing		.65%	3.2	Visual
222.	Radius missing		.65%	3.2	Visual
223.	Finish improper		.65%	3.2	Visual
224.	Evidence of poor workmanship		.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46474B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.4	Body				8838089
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Flash hole missing		100%	3.2	Gage
2.	Crack or split		100%	3.2	Visual
3.	Hydrostatic		100%	3.3	4.4.3.1
Major					
101.	Pitch diameter of thread		.40%	3.2	Gage
102.	Length of thread, min.		.40%	3.2	Gage
103.	Minor diameter of thread, max.		.40%	3.2	Gage
104.	Wall thickness, min.		.40%	3.2	Gage
105.	Thread stripped or otherwise damaged		.40%	3.2	Visual
Minor					
201.	Length of thread, max.		.65%	3.2	Gage
202.	Location of first row of holes		.65%	3.2	Gage
203.	Inside diameter, min.		.65%	3.2	Gage
204.	Depth of counterbore, max.		.65%	3.2	Gage
205.	Diameter of counterbore		.65%	3.2	Gage
206.	Total length, max.		.65%	3.2	Gage
207.	Plug gage fails to pass completely through body		.65%	3.2	Gage
208.	Body fails to pass through required ring gage		.65%	3.2	Gage
209.	Concentricity of inside diameter with pitch diameter of thread		.65%	3.2	Gage
210.	Finish improper		.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46474B (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
211.	Protective coating damaged (base metal exposed)		.65%	3.2	Visual
212.	Evidence of poor workmanship		.65%	3.4	Visual

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4.4.3 Testing

4.4.3.1 Hydrostatic test.-If any primer tubing fails to comply with the applicable requirement, it shall be classed defective and no primer bodies shall be fabricated therefrom. The test shall be performed as specified in 4.5.1.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Test methods and procedures

4.5.1 Hydrostatic test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at a minimum hydrostatic pressure as calculated by the following formula.

$$P = \frac{2\ St}{D} \times .9$$

where:

P = test pressure in pounds per square inch (psi.).
S = minimum specified yield strength, psi.
t = specified minimum wall thickness in inches.
D = specified maximum outside diameter in inches.

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Level C.-Packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-See MIL-A-48078.

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6.2 Submission of Designs for Approval.-See 6.2.3 of MIL-A-48078. Submit equipment designs as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Hydrostatic test.-If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number 1390-A122

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

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4. REMARKS

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